

7.4.25

EE25BTECH11018 - Darisy Sreetej

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Question

The locus of the centre of a circle, which touches the circle is $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis, is given by the equation:

① $x^2 - 6x - 10y + 14 = 0$

② $x^2 - 10x - 6y + 14 = 0$

③ $y^2 - 6x - 10y + 14 = 0$

④ $y^2 - 10x - 6y + 14 = 0$

Solution

Given circle equation can be represented as

$$\|\mathbf{x}\|^2 + 2 \begin{pmatrix} -3 \\ -3 \end{pmatrix}^T \mathbf{x} + 14 = 0 \quad (1)$$

The centre of circle is $\mathbf{c}_1 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ and radius $r = 2$

$$\left(\because f = 14, \quad \mathbf{u} = \begin{pmatrix} -3 \\ -3 \end{pmatrix} \right)$$

Let the centre of the moving circle be $\mathbf{c} = \begin{pmatrix} h \\ k \end{pmatrix}$

As the circle touches Y -axis, Distance of a point from y -axis is given by
radius $R = h$

The equation of moving circle is

$$\|\mathbf{x}\|^2 + 2 \begin{pmatrix} -h \\ -k \end{pmatrix}^T \mathbf{x} + k^2 = 0 \quad (2)$$

From (4.1) and (4.2) , the intersection of curves

$$\|\mathbf{x}\|^2 + 2 \begin{pmatrix} -3 \\ -3 \end{pmatrix}^\top \mathbf{x} + 14 - \|\mathbf{x}\|^2 - 2 \begin{pmatrix} -h \\ -k \end{pmatrix}^\top \mathbf{x} - k^2 = 0 \quad (3)$$

$$2 \begin{pmatrix} h-3 \\ k-3 \end{pmatrix}^\top \mathbf{x} = k^2 - 14 \quad (4)$$

The equation of the common tangent is

$$\begin{pmatrix} h-3 \\ k-3 \end{pmatrix}^\top \mathbf{x} = \frac{k^2 - 14}{2} \quad (5)$$

Distance between the common tangent and centre of given circle equal to it's radius

$$2 = \frac{\left| \begin{pmatrix} h-3 & k-3 \end{pmatrix} \begin{pmatrix} 3 \\ 3 \end{pmatrix} - \frac{k^2-14}{2} \right|}{\sqrt{(h-3)^2 + (k-3)^2}} \quad (6)$$

$$2 = \frac{\left| 3(h-3) + 3(k-3) - \frac{k^2-14}{2} \right|}{\sqrt{(h-3)^2 + (k-3)^2}} \quad (7)$$

$$2\sqrt{(h-3)^2 + (k-3)^2} = 3h + 3k - \frac{k^2}{2} - 11 \quad (8)$$

$$(9)$$

Squaring both sides

$$4((h-3)^2 + (k-3)^2) = (3h + 3k - \frac{k^2}{2} - 11)^2 \quad (10)$$

$$4h^2 + 4k^2 - 24h - 24k + 72 = \frac{k^4}{4} + 9h^2 + 20k^2 + 121 + 18hk - 66h - 66k \quad (11)$$

$$k^4 - 12k^3 + (64 - 12h)k^2 + 72hk + 20h^2 - 168h - 168k + 196 = 0 \quad (12)$$

$$(k^2 - 10h - 6k + 14)(k^2 - 6k - 2h + 14) = 0 \quad (13)$$

The possible general loci are

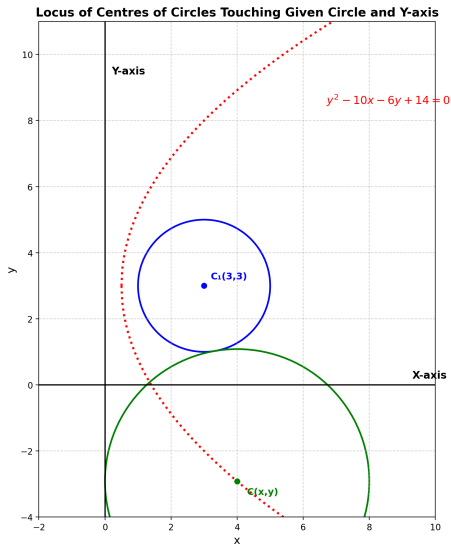
$$y^2 - 10x - 6y + 14 = 0 \quad (14)$$

$$y^2 - 6y - 2x + 14 = 0 \quad (15)$$

Conclusion

From the options , the locus of the centre of the circle is

$$y^2 - 10x - 6y + 14 = 0 \quad (16)$$




```
#include <stdio.h>

void get_circle_params(double* out_data) {
    // Given circle:  $x^2 + y^2 - 6x - 6y + 14 = 0$ 
    // Centre: (3,3), radius = 2
    out_data[0] = 3.0; // x-coordinate
    out_data[1] = 3.0; // y-coordinate
    out_data[2] = 2.0; // radius
}
```

```
import ctypes
import sympy

def find_locus_equation():
    lib = ctypes.CDLL('./code.so')

    double_array_3 = ctypes.c_double * 3
    lib.get_circle_params.argtypes = [ctypes.POINTER(ctypes.c_double)]
    out_data_c = double_array_3()

    lib.get_circle_params(out_data_c)
    c1_x, c1_y, r1 = list(out_data_c) # c1_x=3.0, c1_y=3.0, r1=2.0

    # Symbols: h,k for centre of variable circle
    h, k = sympy.symbols('h k', real=True)
```

```
# We take external tangency and assume  $h \geq 0 \Rightarrow r = h$ 
r = h

lhs = (h - c1_x)**2 + (k - c1_y)**2
rhs = (r + r1)**2

equation = sympy.Eq(lhs, rhs)
locus_expr = sympy.simplify(equation.lhs - equation.rhs)

# Rename symbols to conventional x,y for the returned
# equation
x, y = sympy.symbols('x y', real=True)
final_locus = sympy.simplify(locus_expr.subs({h: x, k: y}))

return sympy.Eq(final_locus, 0)
```

```
if __name__ == '__main__':  
    eq = find_locus_equation()  
    print("Derived locus equation:", eq)det_val = lib.determinant  
        (Y)
```

Python code

```
import sys
sys.path.insert(0, './')
import numpy as np
import matplotlib.pyplot as plt
from call import find_locus_equation

# Get the locus equation
locus_equation = find_locus_equation()
print(f"Locus equation: {locus_equation}")
fig, ax = plt.subplots(figsize=(9, 9))

# Given circle:  $x^2 + y^2 - 6x - 6y + 14 = 0$ 
given_center = (3, 3)
given_radius = 2
given_circle = plt.Circle(given_center, given_radius, color='blue',
                           fill=False, linewidth=2)
ax.add_patch(given_circle)
```

```
# Mark the moving circle centre
ax.plot(x_sample, y_sample, 'go')
ax.text(x_sample + 0.3, y_sample - 0.4, "C(x,y)", color='green',
        fontsize=11, fontweight='bold')

# Mark centre of given circle
ax.plot(*given_center, 'bo')
ax.text(given_center[0] + 0.2, given_center[1] + 0.2, "C(3,3)",
        color='blue', fontsize=11, fontweight='bold')

# Locus (Parabola):
x_vals = np.linspace(0.5, 9, 400)
sqrt_term = np.sqrt(np.maximum(0, 5 * (2 * x_vals - 1)))
y_plus = 3 + sqrt_term
y_minus = 3 - sqrt_term
ax.plot(x_vals, y_plus, 'r:', linewidth=2.5)
ax.plot(x_vals, y_minus, 'r:', linewidth=2.5)
```

Python code

```
# Write the parabola equation neatly near the curve
ax.text(6.7, 8.5, r"$y^2 - 10x - 6y + 14 = 0$", color='red',
        fontsize=13, fontweight='bold')

# Moving circle
x_sample = 4
y_sample = 3 - np.sqrt(5 * (2 * x_sample - 1))
r_sample = x_sample # because circle touches y-axis r = x
moving_circle = plt.Circle((x_sample, y_sample), r_sample, color=
                            'green', fill=False, linewidth=2)
ax.add_patch(moving_circle)

# Formatting , Axes setup
ax.axvline(0, color='black', linestyle='-', linewidth=1.5)
ax.axhline(0, color='black', linestyle='-', linewidth=1.5)
ax.text(9.3, 0.2, "X-axis", fontsize=12, fontweight='bold')
ax.text(0.2, 9.4, "Y-axis", fontsize=12, fontweight='bold')
ax.set_xlim(-2, 10)
```

```
ax.set_ylim(-4, 11)
ax.set_aspect('equal', 'box')
ax.set_xlabel("x", fontsize=13)
ax.set_ylabel("y", fontsize=13)
ax.set_title("Locus of Centres of Circles Touching Given Circle  
and Y-axis", fontsize=14, fontweight='bold')
ax.grid(True, linestyle='--', alpha=0.6)

# Save and show
plt.tight_layout()
plt.savefig('fig.png', dpi=200, bbox_inches='tight')
plt.show()
```